

Searching Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Implement all search algorithms from scratch.
Analyse time and space complexity for each.

Section 1: Linear Search — Internals

★ **Linear Search — Find an Element and Return Index**

★ **Linear Search — Return All Indices of an Element**

★ **Linear Search on a 2D Array (Matrix)**

Linear Search on a Linked List

Find Minimum and Maximum using Single Linear Scan

Section 2: Binary Search — Internals

★ **Binary Search — Iterative Implementation**

★ **Binary Search — Recursive Implementation**

★ **Find First Occurrence of an Element (Lower Bound)**

★ **Find Last Occurrence of an Element (Upper Bound)**

★ **Count Occurrences of an Element using Binary Search**

Find Floor and Ceiling of a Number in Sorted Array

Find Insertion Position of a Target (Search Insert Position)

Section 3: Binary Search Variants

★ **Search in a Rotated Sorted Array**

★ **Find Minimum in a Rotated Sorted Array**

Find Maximum in a Bitonic (Mountain) Array

★ **Search in a Nearly Sorted Array**

Search in an Infinite Sorted Array (Exponential Search)

★ **Find Peak Element using Binary Search**

Find Kth Missing Positive Number using Binary Search

Section 4: Binary Search on Answer

★ **Find Square Root of a Number using Binary Search (Integer)**

★ **Find Nth Root of a Number using Binary Search**

★ **Allocate Minimum Number of Pages (Books) — Binary Search on Answer**

★ **Aggressive Cows Problem — Maximum Minimum Distance**

Minimize Maximum Distance between Gas Stations

Split Array Largest Sum

★ **Capacity to Ship Packages within D Days**

Section 5: Searching in Special Structures

★ **Search in a Row-Wise and Column-Wise Sorted Matrix**

★ **Search in a Sorted Matrix (Staircase Search)**

Find Median of Row-Wise Sorted Matrix using Binary Search

★ **Search in a 2D Matrix (Binary Search on Virtual Array)**

Find a Pair with Sum in a BST

Section 6: Order Statistics

★ **Find Kth Smallest Element in an Array**

★ **Find Kth Largest Element in an Array**

Quick Select Algorithm (Average $O(n)$ Kth Element)

★ **Median of an Unsorted Array**

Find the Kth Smallest Element in a Sorted Matrix

Kth Smallest Element in Two Sorted Arrays

Section 7: Real-World Applications of Searching

★ **Dictionary Word Lookup — Binary Search on Sorted Word List**

★ **Version Control Bug Finder — Binary Search to Find First Bad Commit (Git Bisect Style)**

★ **Library Book Search — Find a Book by ISBN using Binary Search**

E-Commerce Price Filter — Find All Products within a Price Range using Binary Search

★ **Employee Record Lookup — Search by Employee ID in Sorted Database**

Autocomplete Rank Search — Find All Words Starting with a Prefix (Binary Search + Range)

★ **Exam Result Search — Find a Student's Rank using Binary Search on Sorted Score List**